

Assignment:- 2

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Semester: III

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Submission date: 16/09/2025

Short questions:-

① Define Machine Learning?

→ Machine Learning (ML) is a subset of Artificial Intelligence (AI) that allows computers to learn from data and make decisions without being explicitly programmed. Essentially, ML algorithms identify patterns in data and use those patterns to predict or classify new, similar data. This ability to learn from data and improve over time is what makes ML so powerful.

② List any three real-life applications of Machine Learning?

→ The any three real-life applications of ML are as follows:-

1) Recommendation Engine: Services like Netflix, Spotify use ML to suggest movies, shows or songs based on your past viewing and listening habits.

i) NLP: Voice assistants like Siri and Google Assistant responding to commands, translation tool like Google Translate translating between languages and chatbots providing customer support.

iii) Autonomous vehicles: Self-driving cars navigating streets and making driving decisions, and ride-sharing apps like Uber optimizing routes and pricing.

③ What are the main processes involved in Machine Learning?

→ The ML lifecycle is a systematic process of developing and deploying effective ML models, consisting of several stages

- Problem Definition
- Data collection and preparation
- Model selection and Engineering
- Training and validation
- Evaluation and testing
- Deployment and Integration
- Monitoring and Maintenance

4) Differentiate between supervised and unsupervised Learning?

→	Supervised Learning	Unsupervised Learning
i)	uses labeled data (data with known outcomes)	uses unlabeled data (data with no predefined outcomes).
ii)	To predict an outcome based on the training data.	To discover hidden patterns in the data.
iii)	It is learned with a teacher who provides correct answer	Learning on your own by finding similarities or anomalies.

⑤ Give one example of each of classification and regression problems?

→ Classification: Predicting whether an email is spam or not spam. The output is a discrete category.

→ Regression: Predicting the price of a house based on its features (like size, location, and number of bedrooms). The output is a continuous value.

⑥ What is Reinforcement Learning?

→ Reinforcement Learning (RL) is a type of machine learning where an agent learns to make decisions by performing actions in an environment to maximize a cumulative reward. The agent learns through trial and error, much like training a pet with treats for good behaviour. It's not given explicit instructions, but rather a goal to achieve.

⑦ Name two algorithms used in Supervised Learning?

→ The any two algorithms used in Supervised learning are as:-

1) Linear regression: Predicts outcomes by fitting the best straight line through data.

2) Decision Tree: Predicts by splitting data into branches based on rules.

⑧ Name two algorithms used in unsupervised learning?

→ The only two algorithms used in unsupervised learning are:

- 1) k-means clustering: Groups data into k clusters based on similarity.
- 2) PCA (Principal Component Analysis): Reduces data dimensions by keeping the most important features.

⑨ Define training dataset and test dataset?

→ Training dataset

- This is the portion of the data used to train the ML model. The model learns the underlying patterns and relationships from this dataset.
- It's like the set of practice problems you study for an exam.

Test dataset

- This is a separate portion of the data that the model has never seen before. It's used to evaluate the model's performance and check how well it generalizes to new, unseen data. It's like the final exam itself.

⑩ What is the role of a model in Machine Learning?

→ In Machine Learning, a model is the mathematical representation of a real-world process that is the output of the training algorithm.

Essentially, it's the "brain" that has learned from the training data. Its role is to take new input data and make predictions or decisions based on the patterns it has learned.

⑪ State any two advantages of Machine Learning?

→ The advantages are:

i) Automation: ML can automate repetitive and complex tasks, such as data entry, analysis, and decision-making, increasing efficiency and reducing human error.

ii) Pattern Identification: It excels at identifying trends and patterns in large and complex datasets that might be impossible for a human to spot.

⑫ Mention one real-life example of Reinforcement Learning?

→ A great example is self-driving cars. The car's AI acts as the agent, the roads and traffic are the environment and it receives positive rewards for actions like staying in its lane and reaching the destination safely and negative rewards for actions like collisions or breaking traffic laws.

(13) What is the purpose of features selection in machine learning?

→ The purpose of features selection in ML is to choose the most relevant input variables (features) for creating a model. This helps to:

- (a) Improve the model's predictive accuracy.
- (b) Reduce overfitting (when a model performs well on training data but poorly on new data).
- (c) Decrease the computational time and resources needed for training.

(14) List two common challenges in implementing machine learning?

→ The two common challenges in implementing machine learning are as follows:

i) Data quality and quantity
- ML models require large amounts of high-quality, relevant data. Insufficient or dirty data leads to poor performance.

ii) Overfitting:
- The model may learn the training data too well, including its noise and outliers, causing it to fail when exposed to new, unseen data.

Q5) Give one example of machine learning in healthcare.

→ One key example is medical image analysis. ML algorithms are trained to analyze medical scans like X-rays, CT scans, and MRIs to detect diseases such as concave tumors, diabetic retinopathy in eye scans, or signs of Alzheimer's in brain scans, often with accuracy comparable to or exceeding human radiologists.

Long Questions:-

Q1) Explain the various processes involved in machine learning with a neat diagram.

→ Machine learning process involves several key stages that transform raw data into a functional model that can make predictions or decisions.

I) Problem definition

- clearly define the problem that needs to be solved and the objectives to be achieved.

II) Data collection and prediction

→ Gather, cleaning, and labelling data which is crucial for building and training an accurate model.

III) Model selection and Engineering:

- Choosing the right algorithms and building the model architecture based on the problem and data type.

IV) Training and Validation

- Training the model with the prepared data and validating its performance on unseen data to avoid overfitting and underfitting

V) Evaluating and testing:

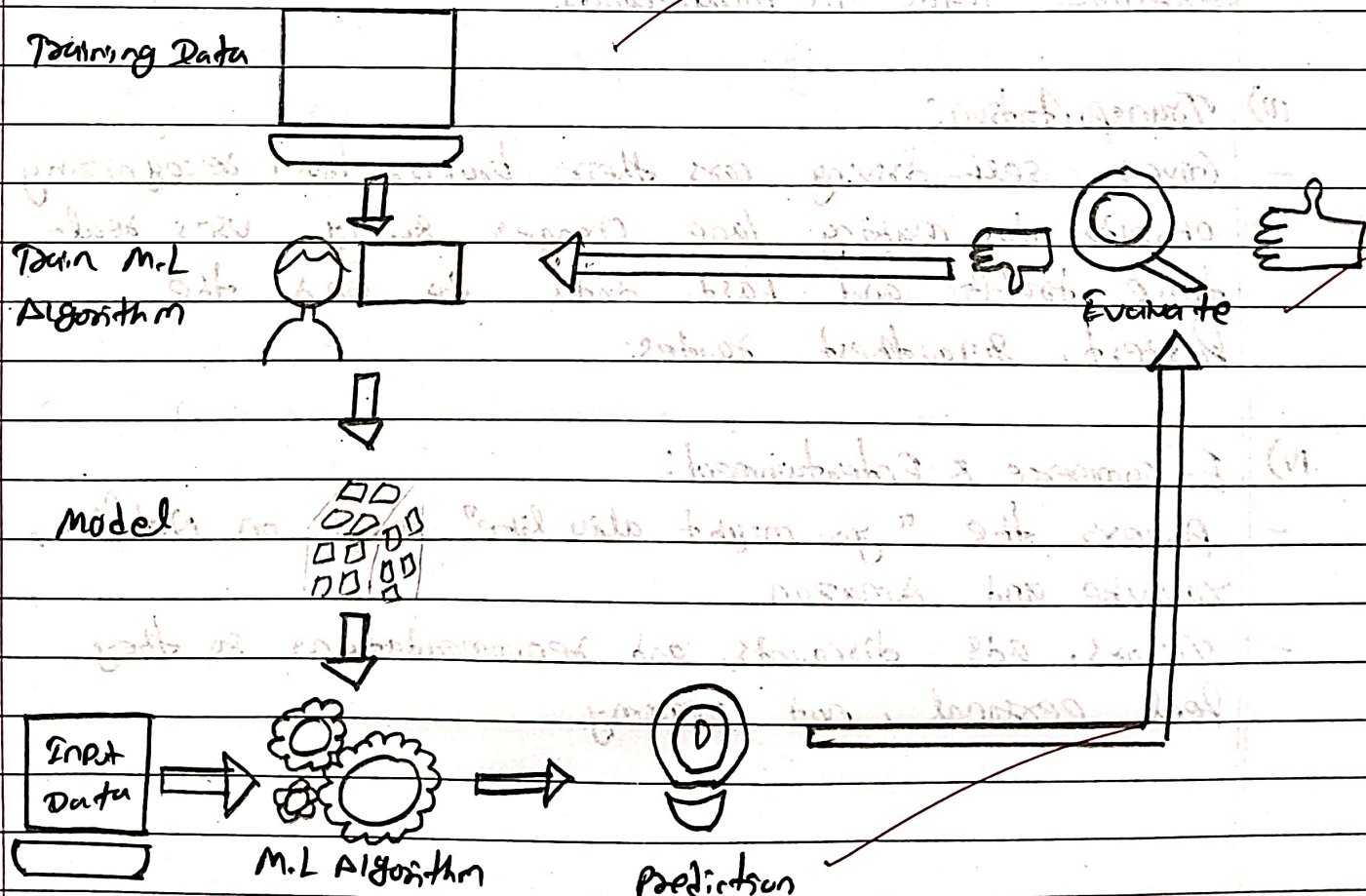
- Rigorously evaluating the model's performance on test data using relevant metrics and making improvements

VI) Deployment and Integration:

- Deploying the model into a live environment, making its predictions accessible through APIs, web apps.

VII) Monitoring and maintenance:

- continuously monitoring the model's performance and retraining it with new data to ensure continued effectiveness.



② Discuss real-life applications of Machine Learning in different domains?

→ ML is transforming numerous industries, here are some key applications.

i) Healthcare:

- Scans - MRIs, CTs, and X-rays to catch tumors or diseases earlier than most doctors can. speed up drug discovery by predicting which compound might actually work before they hit the lab.

ii) Finance

- watches every transactions in real time to catch fraud before it drains your accounts. Runs algorithmic trading bots that can predict market shifts and execute trade in milliseconds.

iii) Transportation:

- Gives self-driving cars their brains - from recognizing objects to making lane changes safely. uses real-time traffic and past data to find the fastest, smoothest routes.

iv) E-commerce & Entertainment:

- Powers the "you might also like" magic on Netflix, Youtube and Amazon.
- Tailors ads discounts, and recommendations so they feel personal, not spammy.

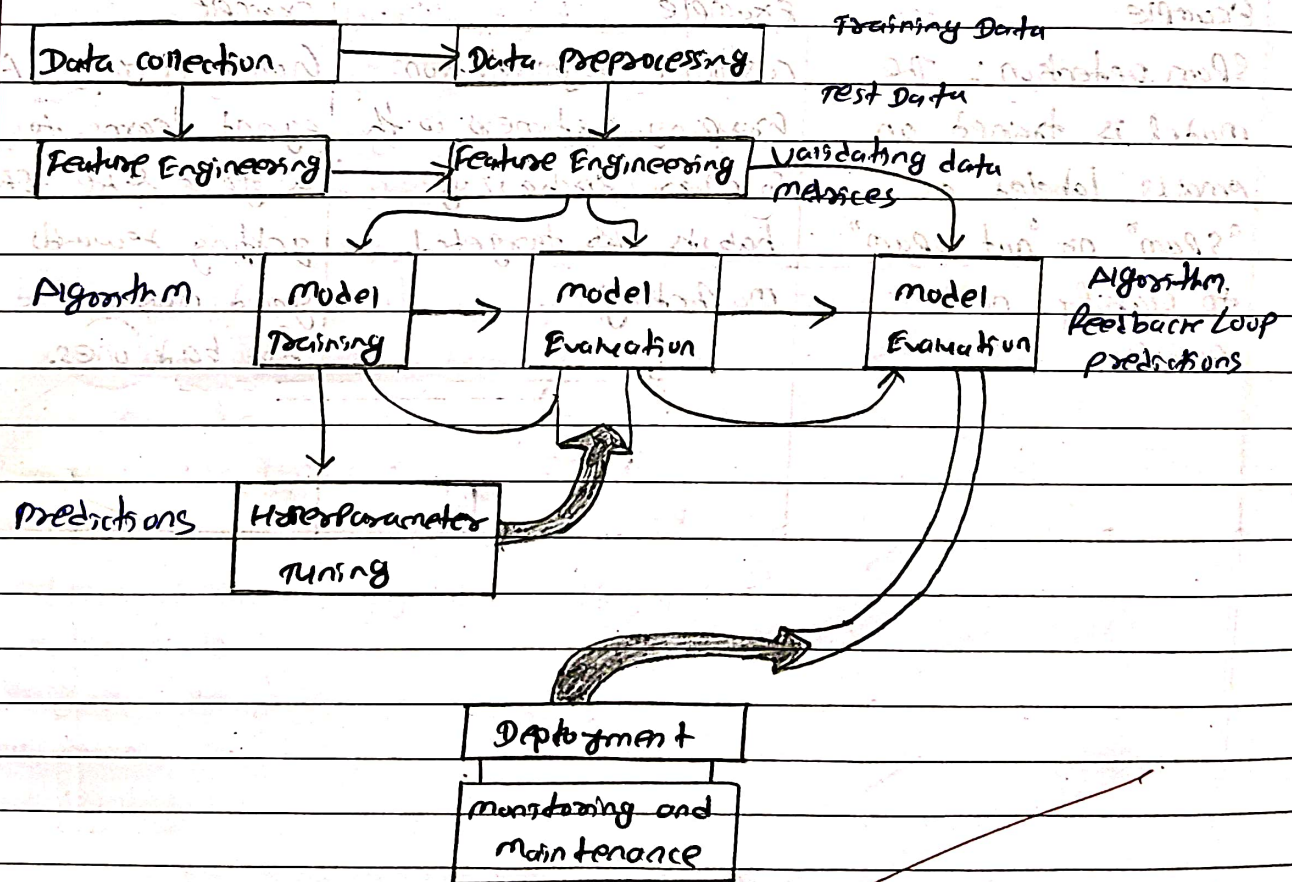
③ compare Supervised, unsupervised and Reinforcement Learnings with examples.

→	Supervised Learning	unsupervised Learning	Reinforcement Learning
i)	Labeled data (input-output pairs)	unlabeled data (input only)	No predefined data; interacts with environment.
ii)	to predict an output based on input data	to discover hidden patterns or structures in data.	to learn the best sequence of action to maximize a reward.
iii)	Direct feedback from the labeled data.	No direct feedback; learns inherent structures.	Feedback through rewards or penalties from action.
iv)	Example Spam detection: The model is trained on emails labeled as "spam" or "not spam" to classify new emails.	Example customer segmentation: grouping customers with similar purchasing habits for targeted marketing.	Example: Game playing: An AI agent learns to play a game like chess by getting rewards for good moves and penalties for bad ones.

④ Describe the workflow of a machine Learning project in detail?

→ A typical ML project follows a structured workflow from conception to deployment.

- i) Define - problem: know what you're solving & set goal.
- ii) collect data: Grab all the relevant info
- iii) Clean & Prep: fix, format and split data
- iv) Explore - spot pattern & insights
- v) Pick model - Choose the right algorithm
- vi) train model - Train it with your data
- vii) Evaluate - test: accuracy & performance
- viii) Deploy - put it into real-world use
- ix) Monitor - keep it fresh & update when needed.



By: Machine Learning workflow.

⑤ Explain Supervised Learning - Discuss its types with suitable examples?

→ It is a type of machine learning where the algorithm learns from data that is labeled. This means for each data input, there is a corresponding correct output. The goal is to learn a mapping function that can predict the output variables (y) from the input data (x). The formula is often represented as $y = f(x)$.

Two types are

i) Classification: The output variable is a category or a class. The model predicts a discrete label.

Ex: Image Recognition: A model is trained on a dataset of thousand image of cat and dogs after training the model can predict whether it's cat or dog.

ii) Regression: The output variable is a continuous, real value. The model predicts a quantity.

Ex: House Price Prediction. A model is trained on a dataset of houses where the features include square footage, no of bedrooms and location, and the label is the price. The trained model can then predict the price of a new house based on the features.

⑥ Explain unsupervised learning. Discuss clustering and dimensionality reduction techniques?

→ Unsupervised Learning is used when you have data with no labels. The goal is to allow the model to explore the data on its own to find meaningful structures or patterns. It's about discovering the underlying distribution of the data.

Techniques

i) Clustering

- This technique involves grouping a set of data points in such a way that points in the same group are more similar to each other than to those in other groups.

ex:

k-means clustering: This algorithm partitions data into k distinct, non-overlapping clusters.

ii) Dimensionality Reduction:

- This technique is used to reduce the number of features in a dataset while retaining as much important information as possible. It helps in simplifying models, reducing storage space, and speeding up training.

ex:

Principal Component Analysis (PCA). It transforms the original features into a new set of uncorrelated features called principal components.

⑦ Explain the concept of Reinforcement Learning with a suitable real-life example?

→ Reinforcement Learning is a goal-oriented learning paradigm where an agent learns to make decisions by performing actions in an environment to maximize a cumulative reward. Unlike supervised learning, there's no correct labeled data. The agent learns from the consequences of its actions through trial and error.

Example: Training a Robot vacuum cleaner

Imagine a new robot vacuum cleaner learning to efficiently clean a room

- Agent: The robot vacuum itself.
- Environment: The room, including furniture, carpet, and obstacles.
- State: The robot's current location, battery level, and sensor reading about nearby objects.
- Action: move forward, turn left, turn right, start suction
- Rewards according to action good and bad

⑧ Describe any three Machine Learning applications in business and industry?

→ ML provides a competitive edge to business across various sectors

i) customer churn prediction

- In telecom, banking, and subscriptions, ML analyzes customer behaviour, contracts, and service history to predict who's likely to leave, allowing proactive retention with targeted offers or better service.

ii) Predictive maintenance in manufacturing:

- unexpected equipment failure can halt production lines, leading to significant financial losses. ML processes sensor data like vibration and temperature to detect any signs of equipment failure, enabling timely maintenance that reduces downtime and costs.

iii) Dynamic Pricing

- In e-commerce, travel, and ride-sharing, ML adjusts prices in real time based on demand, competition, weather and events, helping boost revenue and attract customers.

⑨ Discuss the importance of data preprocessing in the machine learning process?

→ Data preprocessing is the crucial step of cleaning and organizing raw data to make it suitable for a machine learning model. Its importance is often summarized by the principle "Garbage in, garbage out".
 A sophisticated algorithm will produce poor results if fed low-quality data.

Some Important Points:

- **Boosts Accuracy & Performance:** Clean, structured data helps models learn patterns effectively, while noisy data leads to poor predictions.
- **Handles missing & Inconsistent Data:** Techniques like imputation fill gaps, and standardization unifies formats (e.g. USA vs "United States").
- **Makes Data Model-Ready:** Converts categorical values into numerical form so algorithms can process them.
- **Prevents Bias in Results:** Scaling features ensures algorithms don't favor variables with large numeric ranges.

⑩ Explain the advantages, disadvantages and limitations of Machine learning?

→ Advantages

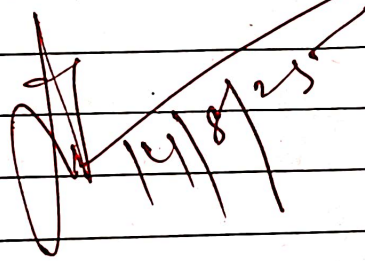
- i) Automation of Tasks - ML reduces the need for manual intervention by automating decision-making and predictions.
- ii) Improved Accuracy over Time - models learn from more data, often becoming more accurate as they're retrained.
- iii) Ability to process Large Data Volumes - can handle massive datasets that humans can't easily analyze.
- iv) Personalization - Enables tailored recommendations, offers, and user experiences.
- v) Early Detection - Identifies patterns or anomalies (e.g. fraud, equipment failure) before issues escalate.

Disadvantages

- i) High Data dependency - performance relies heavily on the quality and quantity of data.
- ii) Complexity - ML models often require advanced expertise.
- iii) Computational cost - some models demand high processing power and storage.
- iv) Potential Bias - Biased training data can lead to unfair or inaccurate predictions.
- v) Interpretability issues - many ML models are "black boxes" that are hard to explain.

Limitations

- i) Not a one-size-fits-all: models trained for one task usually can't be directly applied to another.
- ii) Struggles with unseen scenarios - Poor at predicting events that differ greatly from training data.
- iii) Needs ongoing maintenance - models require retraining to stay relevant as data changes.
- iv) Ethical & legal concerns - privacy issues, misuse risks, and compliance challenges can arise.
- v) Cannot Replace Human Judgement completely - Lacks common sense and contextual understanding.

 14/8/25